

Experiment - 02.

- Aim: Perform the extraction Transformation and loading (ETL) process to construct the database in the SQL server.

- Theory:

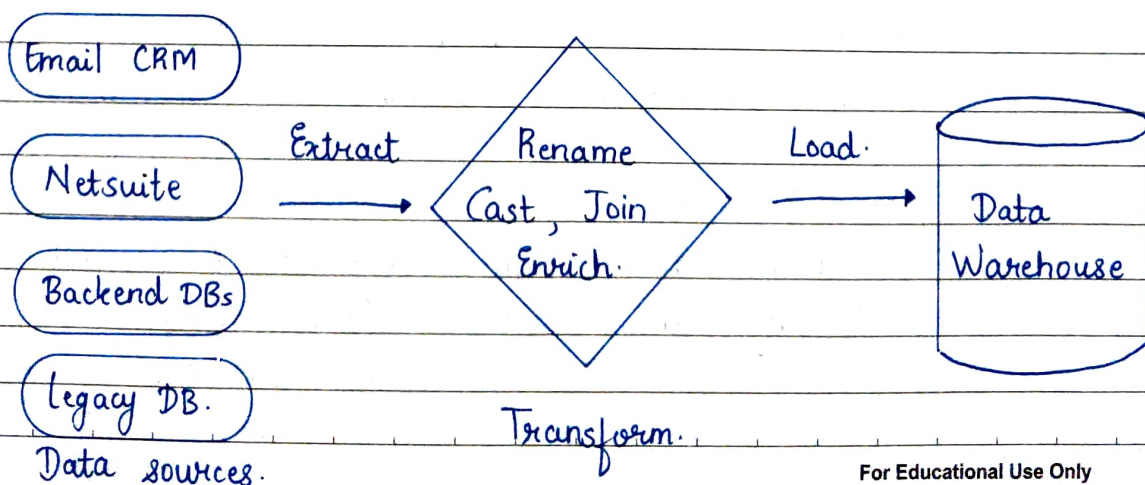
(i). Introduction -

ETL (Extract, Transform, Load) is a data integration process used to collect data from multiple heterogeneous sources, clean, and convert it into a consistent format, and load it into a target system such as Data warehouse in Microsoft SQL Server.

ETL plays a crucial role in:

- Business Intelligence (BI)
- Decision Support Systems (DSS)
- Data warehousing.
- Analytics and reporting.

The ETL process ensures data consistency, quality, integrity, and reliability before it is used for analytical processing.



SO - 1001111111

(2) Components of ETL Architecture -

1. Source systems - Excel files, SQL server databases, Oracle databases, CSV files, Web APIs.
2. Staging area - Temporary storage, raw data storage before transformation.
3. ETL engine - Performs cleaning, filtering, aggregation, validation.
4. Target system - SQL server data warehouse, data mart, reporting database.

(3) ETL Phases -

A. Extraction Phase.

- Extraction involves collecting data from different sources such as : Excel, legacy databases, flat files, ERP systems.
- Objectives of Extraction -
 - Retrieve required data.
 - Minimize source system load.
 - Maintain data integrity.
 - Capture incremental changes (if required)
- Types of Extraction -
 1. Full Extraction - Entire dataset is extracted.
 2. Incremental Extraction - Only changed/new records are extracted.

B Transformation Phase -

Transformation is the most important phase. It converts raw data into structured, clean and standardized format.

Common Transformation operations:-

1. Data cleaning - Remove duplicates, Handle NULL values, correct data types.
2. Data standardization - Date format conversion, currency format normalization, unit conversion.
3. Data integration - Merge multiple source datasets.
4. Data aggregation - Summation, grouping, average calculations.
5. Business rule Application - Profit calculation, Age calculation from date of birth, status classification.

C. Loading phase -

Loading involves inserting transformed data into SQL server target tables.

Types of Loading -

1. Full load. - Entire table is replaced.
2. Incremental load - Only new/updated records inserted.
3. Slowly changing Dimensions (SCD) - Track historical changes.

(4). Practical Methodology to perform ETL in SQL server.

- ETL in SQL server is commonly performed using : SQL server Integration services (SSIS), SQL server management studio (SSMS), TSQL scripts.

Step 1: Prepare the source data - Excel file Sales.xlsx, SQL server database : Adventureworks, csv files.

Step 2: Create Target database in SQL server.

- Open SSMS.
- Create new database.

```
CREATE TABLE DimCustomer (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(100),  
    City VARCHAR(50),  
    Country VARCHAR(50)  
);
```

Step 3: Create SSIS project.

1. Open SQL server Data Tools (SSDT)
2. Create New → Integration Services Project.
3. Add New SSIS Package.

Step 4: Configure Data flow Task.

Inside SSIS -

1. Drag data flow Task.
2. Double click → Data flow Tab.

Step 5: Add source.

- Add excel source / OLE DB Source.
- Configure connection string.
- Select required tables.

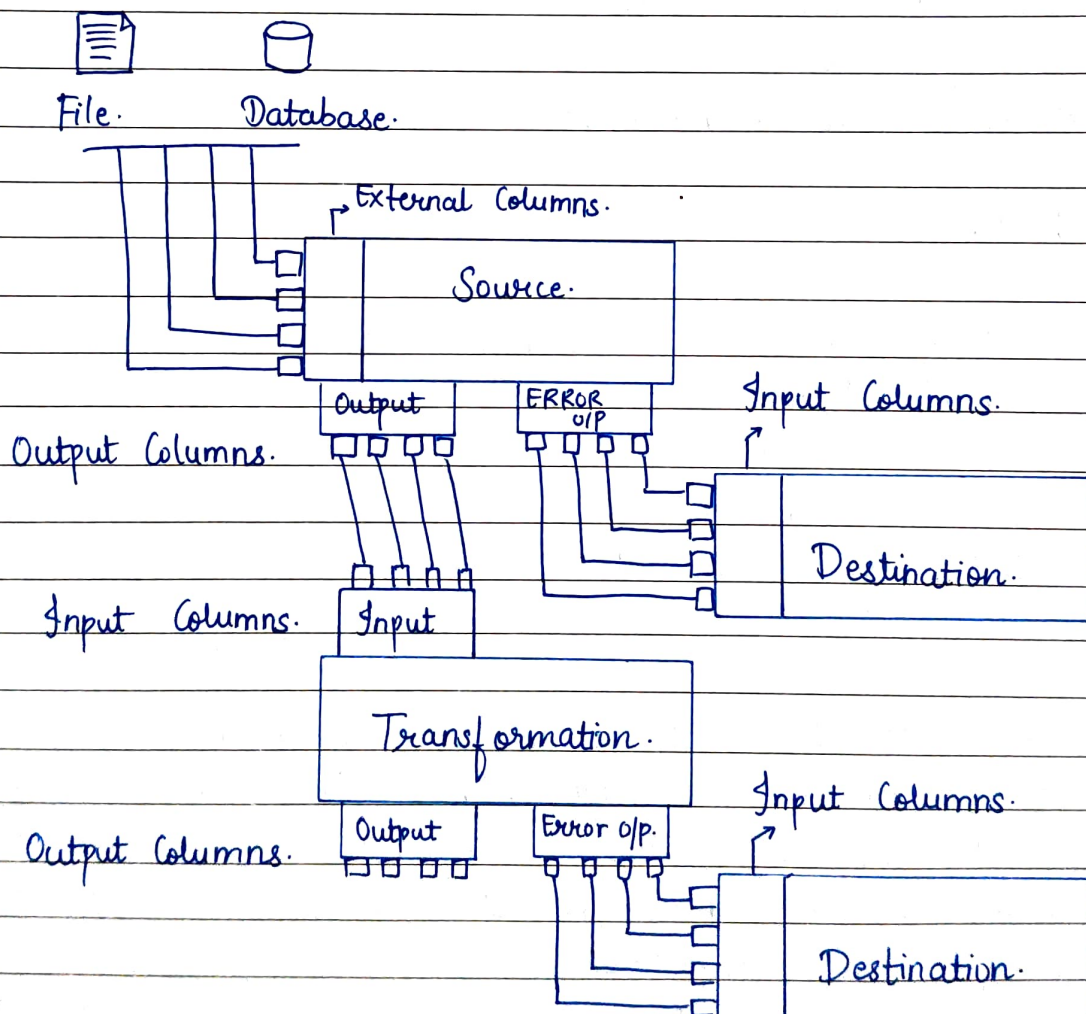
Step 6: Apply Transformations.

Use SSIS transformations such as -

- Data conversion
- Derived column.
- Look up.
- Aggregate.
- Sort
- Conditional split.

Step 7: Load into destination.

- Add OLE DB Destination.
- Map columns.
- Choose table in SQL server.
- Execute Package.



(5). ETL Data Quality considerations -

1. Data validation.
2. Error Handling.
3. Logging and monitoring.
4. Duplicate removal.
5. Referential Integrity checks.

(6). Advantages of ETL in SQL server -

- Centralized data storage.
- Improved reporting performance.
- Better decision making.
- Data consistency.
- Historical data management.

(7). Challenges in ETL.

- Handling large data volumes.
- Schema changes.
- Data quality issues.
- Performance tuning.
- Maintaining incremental loads.

* CONCLUSION: Thus ETL is not merely a data movement process but a data refinement and value creation mechanism in modern information systems. Hence through above methodology ETL to SQL server was completed.

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